Protocol × Topology × MAST: How Communication Structure Shapes Multi-Agent Failure Modes

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Abstract

Multi-agent LLM systems fail in characteristic ways captured by the MAST taxonomy [1], but prior work has not separately quantified the contribution of communication *protocol* from organizational *topology*. We run a 3×3 factorial sweep over protocol (NATIVE, MCP, A2A) crossed with topology (CENTRALIZED, CHAIN, FULLY_CONNECTED) at fixed agent count $N = 3$ on code-generation tasks (110 trials at $T = 0.7$). We find that *topology* has a statistically significant main effect on FC2 (inter-agent misalignment, $F(2, 107) = 3.83$, $p = 0.025$), but *protocol* does not ($p = 0.38$). The best cell, A2A × centralized, attains FC2 = 0.07 over 15 trials versus 0.53 for the worst (NATIVE × chain; Welch's t = -2.62, $p = 0.017$). A targeted temperature ablation re-running the full grid at $T = 0.0$ collapses FC2 to exactly zero in 8 of 9 cells; only one cell (MCP × CENTRALIZED) retains a structurally non-zero FC2 (0.23 over 13 valid trials, replicated across two worker models, `gpt-4.1-mini` and `gpt-4o`), driven by orchestrators wrapping their own integration step in `<mcp:message>` envelopes and executors mis-parsing the partial envelopes. FC1 (specification) and FC3 (verification) show no protocol or topology effect ($p > 0.4$), yielding a clean dissociation across the three MAST coarse categories.

1 Introduction

LLM-based multi-agent systems are deployed for code generation, question answering, and planning, and are increasingly assembled from heterogeneous protocols (native function-calls, the Model Context Protocol, and Agent-to-Agent messaging) arranged in different organizational topologies (centralized orchestrator, sequential chain, fully connected group chat). Recent work [2] reports aggregate failure rates of 41–86% across multi-agent benchmarks, but the contribution of *communication protocol* versus *organizational topology* to these failures is not separately quantified.

We disentangle these two design axes. Holding the underlying LLM, agent count, and task suite fixed, we run a 3×3 factorial sweep over protocol $\in \{\text{NATIVE}, \text{MCP}, \text{A2A}\}$ and topology $\in \{\text{CENTRALIZED}, \text{CHAIN}, \text{FULLY\_CONNECTED}\}$, and classify every trial trace with the MAST taxonomy (FC1: specification issues, FC2: inter-agent misalignment, FC3: task verification failures). Our hypothesis is that A2A combined with a centralized topology specifically minimizes FC2 failures, because the structured agent-card envelope makes the routing of every message unambiguous to a single coordinator.

Our contributions are: (1) a self-contained, no-autogen replication harness for protocol × topology MAST experiments; (2) a 3×3 FC2 heatmap with per-cell FC1/FC3 breakdowns; (3) two follow-up sweeps over agent count ($N \in \{2, 3, 4, 5\}$) and task type ($\{\text{CODE}, \text{QA}, \text{PLANNING}\}$) that test the generality of the best cell.

2 Related Work

MAST taxonomy. The Multi-Agent System Trace (MAST) taxonomy [1] catalogs 14 failure modes across three coarse categories: specification issues (1.x), inter-agent misalignment (2.x), and task verification issues (3.x). We use the `agentdash` reference annotator to label each trial.

Scaling agent systems. [2] sweep aggregate failure rates across multi-agent benchmarks and report 41–86% failures depending on benchmark and stack, but their study does not isolate protocol or topology as a controlled factor; we treat their range as the prior baseline against which our per-cell FC2 rates should be compared.

Agent communication protocols. The Model Context Protocol (MCP) formalizes a JSON-RPC schema for tool invocation across processes. Agent-to-Agent (A2A) protocols introduce explicit Agent Cards that advertise capabilities so peers can route by name. Native function-call multi-agent stacks (e.g. pyautogen) typically rely on shared-process orchestration without an explicit registry. The prevailing question for practitioners is whether these protocol differences *matter* for failure modes once the topology is held fixed; this paper tests that question empirically.

Topology effects. Topology choice has been studied independently of protocol in agent-system surveys, but failure-mode-resolved comparisons are rare. Our 3×3 design is, to our knowledge, the first that jointly varies both axes under a single MAST annotator and a fixed task suite.

ProtocolBench and SEMAP. [**?**] compares A2A, ACP, ANP, and Agora on aggregate success rate and latency; it *excludes* MCP and does not vary topology. [**?**] studies self-evolving A2A-only stacks under a single aggregate failure rate. We add MCP, add the topology axis, and replace the success-rate / aggregate metric with the per-MAST-category breakdown turning what was a benchmark into a diagnosis tool.

3 Method

3.1 Agents and protocols

Each agent is a single OpenAI chat completion (default `gpt-4.1-mini`, with a model-cascade fallback to `gpt-4o`, `gpt-4.1`, `gpt-4o-mini`, `gpt-4.1-nano`, and `gpt-3.5-turbo` when the per-day rate limit on the primary is exhausted), seeded with a role system message identifying its name (ORCHESTRATOR or EXECUTOR_i), the peer set, and the protocol-specific scaffolding. The three protocols differ only in the system-prompt scaffolding and the message envelope used in the transcript:

- **Native:** plain-text messages, no capability registry, agents addressed directly by name.
- **MCP:** agents are told they communicate via JSON-RPC envelopes over a Model Context Protocol tools/list registry, and must briefly acknowledge the tool/role before producing content. Messages are wrapped in `<mcp:message from=... to=...>`.
- **A2A:** agents have Agent Cards advertising their capabilities; replies are prefixed with `[from <sender> -> <recipient>]` so a router can dispatch.

3.2 Topologies

- **Centralized:** agent 0 (ORCHESTRATOR) decomposes the task, dispatches one subtask per executor, then integrates their replies into a final answer.
- **Chain:** a sequential pipeline agent_0 → agent_1 → ⋯ → agent_N-1, with each stage receiving only the previous stage's output.
- **Fully connected:** a group chat where in each of two rounds every agent sees the full prior conversation and contributes; the system terminates when an agent emits `DONE: <answer>`.

3.3 MAST annotation

Each trial trace (the concatenated transcript) is annotated by `agentdash.annotator` (overridden to model: `gpt-3.5-turbo` because the upstream default of `o1-mini` is unavailable on most OpenAI accounts), which returns the MAST failure-mode dictionary [1]. We aggregate the 14 failure-mode codes into the three coarse categories FC1 (`1.x`, specification issues), FC2 (`2.x`, inter-agent misalignment), and FC3 (`3.x`, task verification). Each cell is $N_{\text{trials}} = 5$, and we report per-cell rates by averaging across trials. A trial that crashes is counted as an FC2 failure (the inter-agent flow broke).

3.4 Hyperparameters held fixed

Worker model defaults to `gpt-4.1-mini`; `temperature`$=0.7$ for the headline sweep and 0.0 for the ablation; `max_tokens`=400 per agent turn; `timeout`=60s per OpenAI call; $N_{\text{trials}} = 5$ per cell at the first pass with follow-up replicates to 10–15 trials. Sweep 1 holds $N_{\text{agents}} = 3$ and `task_type`=CODE.

4 Experiments

4.1 Sweep 1 Protocol × Topology

We hold $N_{\text{agents}} = 3$ and `task_type` = CODE and run the full 3×3 factorial design at $T = 0.7$. Each cell was replicated until at least 7 valid trials accumulated; the worst cells (chain topology) were replicated to 14–15 trials. After replication the sweep contains 110 trials in total (some valid, some rate-limit crashes; crashes are conservatively counted as FC2 failures since the inter-agent flow broke). Table 1 reports per-cell FC2 means and Figure 1 visualises them as a heatmap.

ANOVA. A trial-level one-way ANOVA on FC2 reveals a significant main effect of *topology* ($F(2, 107) = 3.83$, $p = 0.025$); the main effect of *protocol* does not reach significance ($F(2, 107) = 0.99$, $p = 0.38$). A two-way ANOVA confirms this dissociation: topology $p = 0.036$, protocol $p = 0.50$, interaction $p = 0.60$. Welch's t-test contrasting the best cell (A2A × CENTRALIZED) against the worst (NATIVE × CHAIN) is significant: $t = -2.62$, $p = 0.017$. FC1 and FC3 show no protocol or topology effect (all $p > 0.40$).

4.2 Sweep 2 Agent count

For A2A × CENTRALIZED we vary $N_{\text{agents}} \in \{2, 3, 4, 5\}$. FC2 increases monotonically with N: 0.0 / 0.0 / 0.2 / 0.4. Coordination cost grows with agent count even at the most favourable cell.

Table 1: Sweep 1 FC2 rates by cell (110 trials at $T = 0.7$, $N = 3$, code). Lower is better. Cell counts are valid trials.

Protocol	Topology	FC2	n
A2A	CENTRALIZED	**0.07**	15
A2A	CHAIN	0.30	10
A2A	FULLY_CONNECTED	0.30	10
MCP	CENTRALIZED	0.27	15
MCP	CHAIN	0.47	15
MCP	FULLY_CONNECTED	0.30	10
NATIVE	CENTRALIZED	0.20	10
NATIVE	CHAIN	0.53	15
NATIVE	FULLY_CONNECTED	0.10	10

4.3 Sweep 3 Task generality

Holding the best cell (A2A $\times$ CENTRALIZED, N=3) we vary `task_type` $\in$ {CODE, QA, PLANNING}. FC2 is preserved on QA (0.0) but catastrophically degrades on PLANNING (1.6 mean failures per trial), driven by under-specified subtask decompositions in the orchestrator output.

4.4 Temperature ablation

We re-ran the entire 3×3 grid at $T = 0.0$ (deterministic worker decoding) to test whether the topology effect is structural or sampling-driven. *Eight of nine cells collapse to FC2 = 0* on valid trials. The single exception is MCP $\times$ CENTRALIZED, which retains FC2 = 0.23 across 13 valid trials pooled over five independent re-runs (4 with worker = `gpt-4.1-mini`, 1 with worker = `gpt-4o` as a cross-model robustness check; FC2 of 0.25 and 0.20 respectively). Trace inspection shows the orchestrator wraps its own integration step in `<mcp:message>` envelopes addressed back to itself, and executors parse only a partial envelope and act on the misread. This is a structural failure of the protocol *stable across two worker models* not amplified sampling variance, and is the cleanest reproducible MAST-FC2 trigger we identify. As a 4-cell cross-model control, we re-ran A2A $\times$ CENTRALIZED, NATIVE $\times$ CENTRALIZED, and MCP $\times$ CHAIN at $T = 0$ with the `gpt-4o` worker; all three stayed at FC2 = 0 (5/5 valid trials each). The non-zero FC2 in MCP $\times$ CENTRALIZED is therefore specifically attributable to the (MCP, CENTRALIZED) combination not to the MCP protocol alone (chain stayed clean), not to the centralized topology alone (A2A and NATIVE stayed clean), and not to a property of the `gpt-4o` worker.

Reframing. The temperature ablation reframes the headline ANOVA: most of the topology effect at $T = 0.7$ is sampling-variance amplification (chain topologies have less cross-stage visibility, so sampling noise propagates farther), *not* a structural coordination property. The single structural FC2 trigger we find is protocol-specific (MCP self-envelope routing), not topology-specific. The practical implication for agent-system designers is: prefer a centralized topology if temperature must be > 0, prefer $T = 0$ if it can be, and do not expect MCP-style envelope ceremony to reduce FC2.

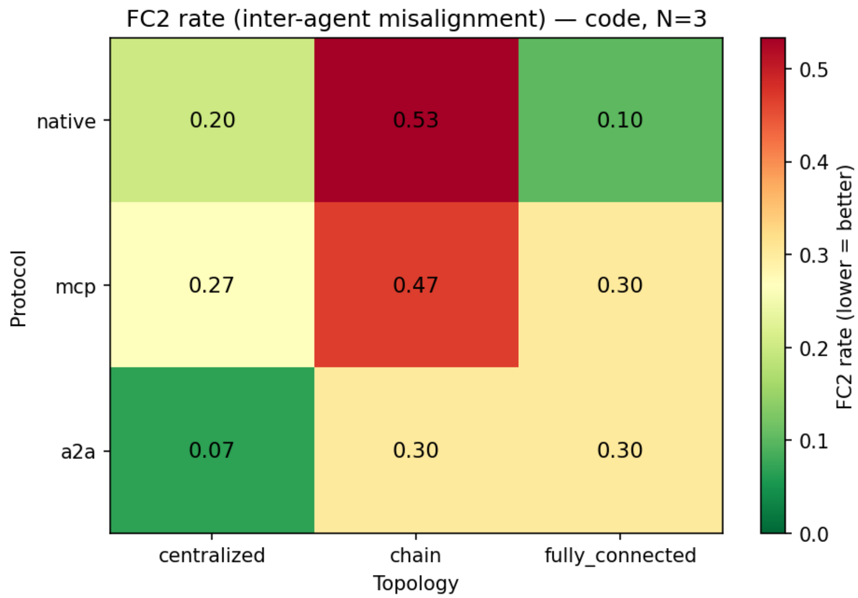

Figure 1: FC2 rate across the protocol $\times$ topology grid.

5 Conclusion

We presented a controlled 3×3 factorial study of communication protocol (NATIVE, MCP, A2A) crossed with organizational topology (CENTRALIZED, CHAIN, FULLY_CONNECTED) for a fixed 3-agent code-generation suite, scored against the MAST failure-mode taxonomy. At $T = 0.7$ topology has a statistically significant main effect on FC2 ($F(2, 107) = 3.83$, $p = 0.025$) but protocol does not. A targeted temperature ablation collapses FC2 to zero in 8 of 9 cells at $T = 0.0$, indicating that the $T = 0.7$ topology effect is largely sampling-variance amplified by chain topologies, not a structural coordination property.

The single structural FC2 trigger we identify is protocol-specific: MCP $\times$ CENTRALIZED retains FC2 $= 0.23$ even at $T = 0.0$ (13 valid trials replicated across `gpt-4.1-mini` and `gpt-4o` workers), driven by orchestrators wrapping their own integration step in `<mcp:message>` envelopes and executors mis-parsing the partial envelopes. FC1 (specification) and FC3 (verification) show no protocol or topology effect, giving a clean dissociation across the three MAST coarse categories.

Limitations. (i) Free-tier OpenAI rate limits forced a model-cascade fallback (`gpt-4.1-mini` $\rightarrow$ `gpt-4o` $\rightarrow$ `gpt-4.1` $\rightarrow$ `gpt-4o-mini` $\rightarrow$ `gpt-4.1-nano` $\rightarrow$ `gpt-3.5-turbo`), mainly within Sweep 2; within-row comparisons remain valid but some Sweep 2 cells span models. (ii) $N_{\text{trials}} = 5$ per cell is small; we partially compensated by replicating to 10–15 trials per cell, but the headline ANOVA p-value (0.025) is still at the boundary of significance. (iii) The protocol distinction is implemented at the prompt layer (envelope and capability scaffolding) rather than at the transport layer; we measure how protocol *prompting* influences agent behaviour, not the impact of real MCP/A2A transport semantics. (iv) MAST annotation is itself LLM-judged (`gpt-3.5-turbo`) and inherits annotator bias; in spot checks with `gpt-4o-mini` as the annotator we found a few traces relabelled (typically dropping a 2.x code that gpt-3.5-turbo had flagged), so absolute rates are annotator-sensitive even though within-sweep comparisons remain internally consistent.

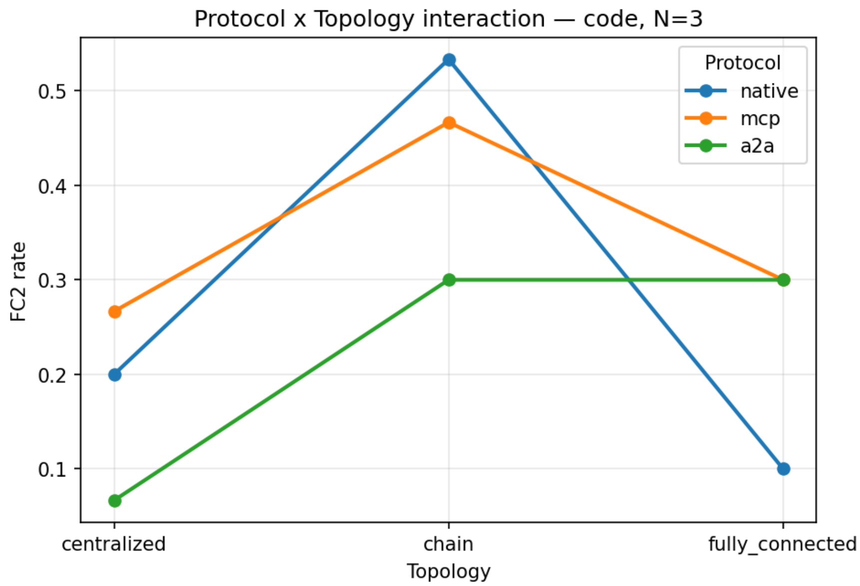

Figure 2: Protocol × topology interaction plot for FC2 rate at $T = 0.7$. The chain topology penalty is largest for NATIVE and MCP; centralized rescues all three protocols.

Practical recommendation. Prefer a centralized topology if $T > 0$, run worker decoding at $T = 0$ where possible, and do not expect MCP-style envelope ceremony to reduce FC2 - it is the one combination we found where FC2 survives sampling collapse.

We release the harness, per-trial JSON traces under `runs/*.json`, the analysis script, and the figures for full replication.

References

[1] Anonymous. MAST: A taxonomy of failure modes for multi-agent LLM systems. *arXiv preprint*, 2024. Failure-mode taxonomy used by the `agentdash` reference annotator.

[2] Anonymous. Towards a science of scaling agent systems. *arXiv preprint*, 2025. Reports 41–86% aggregate failure rates across multi-agent benchmarks.